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Overview of EXIST 2024 — Learning with Disagreement for Sexism Identification and Characterization in Tweets and Memes

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Abstract. In recent years, the rapid increase in the dissemination of offensive and discriminatory material aimed at women through social media platforms has emerged as a significant concern. This trend has had adverse effects on women’s well-being and their ability to freely express themselves. The EXIST campaign has been promoting research in online sexism detection and categorization in English and Spanish since 2021. The fourth edition of EXIST, hosted at the CLEF 2024 conference, consists of three groups of tasks, which are a continuation of EXIST 2023: *sexism identification*, *source intention identification*, and *sexism categorization*. However, while EXIST 2023 focused on processing tweets, the novelty of this edition is that the three tasks are also applied to memes, resulting in a total of six tasks. The “learning with disagreement” paradigm is adopted to address disagreements in the labelling process and promote the development of equitable systems that are able to learn from different perspectives on the sexism phenomena. The 2024 edition of EXIST has exceeded the success of previous editions, with the participation of 57 teams submitting 412 runs. This lab overview describes the tasks, dataset, evaluation methodology, participant approaches and results.

Keywords: sexism identification · sexism categorization · learning with disagreement · memes · data bias

1 Introduction

EXIST (sEXism Identification in Social neTworks) is a series of scientific events and shared tasks on sexism identification in social networks. The editions of

2021 and 2022 [36,37], celebrated under the umbrella of the IBERLEF forum, were the first in proposing tasks focusing on identifying and classifying online sexism in a broad sense, from explicit and/or hostile to other subtle or even benevolent expressions. The 2023 edition [33] took place as a CLEF Lab and added a third task consisting in determining the intention of the author of sexist messages with the aim of raising awareness against sexism. Additionally, the main novelty of the 2023 edition was the adoption of the “Learning with Disagreements” (LwD) paradigm [47] for the development of the dataset and for the evaluation of the systems. In the LwD paradigm, models are trained to handle and learn from conflicting or diverse annotations so that different annotators’ perspectives, biases, or interpretations are taken into account. This approach fits the findings of our previous work that showed that the perception of sexism is strongly dependent on the demographic and cultural background of the individual. Adopting this paradigm was a distinguishing feature in comparison to the SemEval-2023 Shared Task 10: “Explainable Detection of Online Sexism” [18].

EXIST 2024,¹ organised also as a CLEF Lab, aims to continue contributing datasets and tasks that help developing applications to combat sexism on-line, as a form of hate on-line. This edition embraces also the LwD paradigm and, as novelty, incorporates three new tasks that center around memes. Memes are images that are spread rapidly by social networks and Internet users. While by nature memes are humorous, there is a growing tendency to use them for harmful purposes, as an strategy to conceal hate speech by combining stylistic devices of humour [5], since people tolerate humorously communicated prejudices better than explicit disrespectful remarks [8,19,24]. Thus, memes contribute to spreading derogatory humour and to strengthen preexisting prejudices and maintaining hierarchies between social groups [14]. As Gasparini et al. indicate [12], misogyny and sexism against women are widespread attitudes within the social media communities, reinforcing age-old patriarchal establishments of baseless name-calling, objectifying their appearances, and stereotyping gender roles. By including sexist memes in the EXIST 2024 dataset, we aim to encompass a broader spectrum of sexist manifestations in social networks and to contribute to the development of automated multimodal tools capable of detecting harmful content targeting women.

Meme detection has also been the focus of other competitions. The SemEval-2022 Task 5: Multimedia Automatic Misogyny Identification [11] focused on the detection of misogynous memes on the web in English and proposed two tasks: recognising whether a meme is misogynous or not and recognising types of misogyny in memes. The Overview of Shared Task on Multitask Meme Classification - Unraveling Misogynistic and Trolls in Online Memes [7] consisted in classifying misogynistic content and troll memes, focusing specifically on memes in Tamil and Malayalam languages. The originality of EXIST lies in that the languages addressed are English and Spanish, it introduces also the task on source intention recognition and it adopts the LwD paradigm.

¹ <http://nlp.uned.es/exist2024/>. Accessed 28 May 2024.

In the following sections, we provide comprehensive information about the tasks, the dataset, the evaluation methodology, the results and the different approaches of the systems that participated in the EXIST 2024 Lab. The competition features six distinct tasks: sexism identification, source intention classification, and sexism categorization both in tweets and in memes. A total of 148 teams from 32 different countries registered to participate. Ultimately, we received 412 results from 57 teams.

2 Tasks

The 2024 edition of EXIST feature 6 tasks, which are described below. The languages addressed are English and Spanish and the datasets are collections of tweets and memes. For the tasks on memes, all the partitions of the dataset are new, whereas for the tasks on tweets we employ the EXIST 2023 dataset.

2.1 Task 1: Sexism Identification in Tweets

This is a binary classification task where systems must decide whether or not a given tweet expresses ideas related to sexism in any of the three forms: it is sexist itself, it describes a sexist situation in which discrimination towards women occurs, or criticizes a sexist behaviour. The following statements from the dataset show examples of sexist and not sexist messages, respectively.

- (1) **Sexist.** *It's less of #adaywithoutwomen and more of a day without feminists, which, to be quite honest, sounds lovely.*
- (2) **Not sexist.** *Just saw a woman wearing a mask outside spank her very tightly leashed dog and I gotta say I love learning absolutely everything about a stranger in a single instant.*

2.2 Task 2: Source Intention in Tweets

This task aims to categorize the message according to the intention of the author. We propose the following ternary classification of tweets:

- **Direct** sexist message. The intention is to write a message that is sexist by itself or incites to be sexist, as in:
 - (3) *A woman needs love, to fill the fridge, if a man can give this to her in return for her services (housework, cooking, etc.), I don't see what else she needs.*
- **Reported** sexist message. The intention is to report and share a sexist situation suffered by a woman or women in first or third person, as in:
 - (4) *I doze in the subway, I open my eyes feeling something weird: the hand of the man sat next to me on my leg #SquealOnYourPig.*
- **Judgemental** message. The intention is to condemn sexist situations or behaviours, as in:
 - (5) *As usual, the woman was the one quitting her job for the family's welfare...*

2.3 Task 3: Sexism Categorization in Tweets

Many facets of a woman's life may be the focus of sexist attitudes including domestic and parenting roles, career opportunities, sexual image, and life expectations, to name a few. According to this, each sexist tweet must be assigned one or more of the following categories:

- **Ideological and inequality.** It includes messages that discredit the feminist movement. It also includes messages that reject inequality between men and women, or present men as victims of gender-based oppression.
 - (6) *#Feminism is a war on men, but it's also a war on women. It's a war on female nature, a war on femininity.*
- **Stereotyping and dominance.** It includes messages that express false ideas about women that suggest they are more suitable or inappropriate for certain tasks, and somehow inferior to men.
 - (7) *Most women no longer have the desire or the knowledge to develop a high quality character, even if they wanted to.*
- **Objectification.** It includes messages where women are presented as objects apart from their dignity and personal aspects. We also include messages that assume or describe certain physical qualities that women must have in order to fulfill traditional gender roles.
 - (8) *No offense but I've never seen an attractive african american hooker. Not a single one.*
- **Sexual violence.** It includes messages where sexual suggestions, requests or harassment of a sexual nature (rape or sexual assault) are made.
 - (9) *I wanna touch your tits..you can't imagine what I can do on your body.*
- **Misogyny and non sexual violence.** It includes expressions of hatred and violence towards women.
 - (10) *Domestic abuse is never okay... Unless your wife is a bitch.*

2.4 Task 4: Sexism Identification in Memes

As in Task 1, this is a binary classification consisting on deciding whether or not a meme is sexist. Figure 1 shows examples of sexist and non sexist memes.

2.5 Task 5: Source Intention in Memes

As in Task 2, this task aims to categorize the meme according to the intention of the author. However, in this task systems should only classify memes in two classes: DIRECT or JUDGEMENTAL, as shown in Fig. 2.

2.6 Task 6: Sexism Categorization in Memes

This task aims to classify sexist memes according to the categorization provided for Task 3: (i) ideological and inequality, (ii) stereotyping and dominance, (iii) objectification, (iv) sexual violence and (v) misogyny and non-sexual violence. Figure 3 shows one meme of each category.



Fig. 1. Examples of sexist and non-sexist memes.

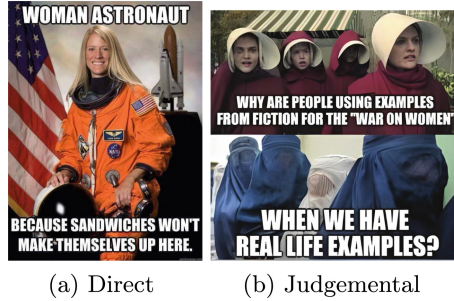


Fig. 2. Examples of direct and judgemental memes.

3 Dataset

The EXIST 2024 dataset comprises two types of data: the tweets from the EXIST 2023 dataset and a completely new dataset of memes. Here, we briefly describe the process followed to curate the meme dataset. More details – including bias considerations, the annotation process, quality experiments, and inter-annotator agreement – can be found in the extended overview [32]. In contrast, Plaza et al. [33] provide a detailed description of the tweet dataset.

Since we adopt the LwD paradigm, we provide all labels assigned by the different annotators to allow systems to learn from conflicting and subjective information. This paradigm not only proved to improve the systems’ accuracy, robustness and generalizability, but also helped to mitigate bias.

3.1 Data Sampling

We first curated a lexicon of terms and expressions leading to sexist memes. The set of seeds encompasses diverse topics and contains 250 terms, with 112 in English and 138 in Spanish.



Fig. 3. Examples of memes from the different sexist categories.

The terms were used as search queries on Google Images to obtain the top 100 images. Rigorous manual cleaning procedures were applied, defining memes and ensuring the removal of noise such as textless images, text-only images, ads, and duplicates. The final set consists of more than 3,000 memes per language.

Since the proportion of memes per term was heterogeneous, we discarded the most unbalanced seeds and made sure that all seeds have at least five memes. To avoid introducing selection bias, we randomly selected memes, ensuring the appropriate distribution per seed. As a result, we have 2,000 memes per language for the training set and 500 memes per language for the test set.

3.2 Labeling with Disagreements

We have considered some sources of “label bias”, that may be introduced by the socio-demographic differences of the persons that participate in the annotation process, but also when more than one possible correct label exists or when the decision on the label is highly subjective. In order to mitigate label bias, we consider two sociodemographic parameters: gender (MALE/FEMALE) and age (18–22/23–45/+46 y.o.). Each meme was annotated by 6 annotators selected through the Prolific crowdsourcing platform.² Also, as a new feature in

² No personally identifiable information about the crowd workers was collected. Crowd workers were informed that the tweets could contain offensive information and were allowed to withdraw voluntarily at any time. Full consent was obtained.

the datasets, both of 2023 and 2024, we have requested three additional demographic characteristics of annotators: level of education, ethnicity, and country of residence.

4 Evaluation Methodology and Metrics

As in EXIST 2023, we have carried out a “**soft evaluation**” and a “**hard evaluation**”. The soft evaluation relates to the LwD paradigm and is intended to measure the ability of the model to capture disagreements, by considering the probability distribution of labels in the output as a soft label and comparing it with the probability distribution of the annotations. The hard evaluation is the standard paradigm and assumes that a single label is provided by the systems for every instance in the dataset.

1. **Soft-soft evaluation.** For systems that provide probabilities for each category, we perform a soft-soft evaluation that compares the probabilities assigned by the system with the probabilities assigned by the set of human annotators. The probabilities of the classes for each instance are calculated according to the distribution of labels and the number of annotators for that instance. We use a modification of the original ICM metric (Information Contrast Measure [1]), ICM-Soft (see details below), as the official evaluation metric in this variant and we also provide results for the normalized version of ICM-Soft (ICM-Soft Norm).
2. **Hard-hard evaluation.** For systems that provide a hard, conventional output, we perform a hard-hard evaluation. To derive the hard labels in the ground truth from the different annotators’ labels, we use a probabilistic threshold computed for each task. As a result, for Tasks 1 and 4, the class annotated by more than 3 annotators is selected; for Tasks 2 and 5, the class annotated by more than 2 annotators is selected; and for Tasks 3 and 6 (multilabel), the classes annotated by more than 1 annotator are selected. The instances for which there is no majority class (i.e., no class receives more probability than the threshold) are removed from this evaluation scheme. The official metric for this task is the original ICM, as defined by Amigó and Delgado. We also report a normalized version of ICM (ICM Norm) and F1 (F1_{YES}). In Tasks 1 and 4, we use F1 for the positive class. In Tasks 2, 3, 5 and 6, we use the macro-average of F1 for all classes (Macro F1). Note, however, that F1 is not ideal in our experimental setting: although it can handle multilabel situations, it does not take into account the relationships between classes. In particular, a confusion between not sexist and any of the sexist subclasses, and a confusion between two of the sexist subclasses, are penalized equally.

ICM is a similarity function that generalizes Pointwise Mutual Information (PMI), and can be used to evaluate outputs in classification problems by computing their similarity to the ground truth. The general definition of ICM is:

$$\text{ICM}(A, B) = \alpha_1 \text{IC}(A) + \alpha_2 \text{IC}(B) - \beta \text{IC}(A \cup B)$$

Where $IC(A)$ is the Information Content of the instance represented by the set of features A . ICM maps into PMI when all parameters take a value of 1. The general definition of ICM by [1] is applied to cases where categories have a hierarchical structure and instances may belong to more than one category. The resulting evaluation metric is proved to be analytically superior to the alternatives in the state of the art. The definition of ICM in this context is:

$$ICM(s(d), g(d)) = 2IC(s(d)) + 2IC(g(d)) - 3IC(s(d) \cup g(d))$$

Where $IC()$ stands for Information Content, $s(d)$ is the set of categories assigned to document d by system s , and $g(d)$ the set of categories assigned to document d in the gold standard. The score for a perfect output ($s(d) = g(d)$) is the gold standard Information Content ($IC(g(d))$). The score for a zero-information system (no category assignment) is $-IC(g(d))$. We use these two boundaries for normalisation purposes, truncating to 0 the scores lower than $-IC(g(d))$.

As there is not, to the best of our knowledge, any current metric that fits hierarchical multilabel classification problems in a LwD scenario, we have defined an extension of ICM (ICM-soft) that accepts both soft system outputs and soft ground truth assignments. ICM-soft works as follows: first, we define the Information Content of a single assignment of a category c with an agreement v to a given instance as the probability of instances in the gold standard to exceed the agreement level v for the category c :

$$IC(\langle\{c, v\}\rangle) = -\log_2(P(\{d \in D : g_c(d) \geq v\}))$$

In order to estimate IC , we compute the mean and deviation of the agreement levels for each class across instances, and applying the cumulative probability over the inferred normal distribution.³

Due to the multi-label and hierarchical nature of the classification task, for each classification instance, the gold standard, the system output and their unions ($IC(s(d))$ $IC(g(d))$ and $IC(s(d))Ug(d)$) are sets of category assignments. The union of the assignments (i.e. $s(d)Ug(d)$) is calculated as fuzzy sets, i.e. the maximum values., in order to estimate information content, we apply a recursive function similar to the one described by Amigó and Delgado [1] for assignment sets and avoid the redundant information of parent categories.

$$\begin{aligned} IC\left(\bigcup_{i=1}^n \{\langle c_i, v_i \rangle\}\right) &= IC(\langle\{c_1, v_1\}\rangle) + IC\left(\bigcup_{i=2}^n \{\langle c_i, v_i \rangle\}\right) \\ &\quad - IC\left(\bigcup_{i=2}^n \{\langle \text{lca}(c_1, c_i), \min(v_1, v_i) \rangle\}\right) \end{aligned} \quad (1)$$

where $\text{lca}(a, b)$ is the lowest common ancestor of categories a and b .

³ In the case of zero variance, we must consider that the probability for values equals or below the mean is 1 (zero IC) and the probability for values above the mean must be smoothed. But this is not the case of the EXIST datasets.

Table 1. Runs submitted and teams participating on each EXIST 2024 task.

	Tweets			Memes		
	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6
# Runs	106	77	63	87	36	43
# Teams	46	38	27	41	18	22

5 Overview of Approaches

In this section, we provide a concise overview of the approaches presented at EXIST 2024. For a comprehensive description of the systems, please refer to the participant papers and the extended overview [32]. Although 148 teams from 32 different countries registered for participation, the number of participants who finally submitted results were 57, submitting 412 runs. Teams were allowed to participate in any of the six tasks and submit hard and/or soft outputs. Table 1 summarizes the participation in the different tasks and evaluation contexts.

The evaluation campaign started on March 4, 2024 with the release of the training set. The test set was made available on April 15. The participants were provided with the official evaluation script. Runs had to be submitted by May 10. Each team could submit up to three runs per task.

A wide range of approaches and strategies were used by the participants. Here we summarize the techniques and tools employed. For a comprehensive description, please refer to the extended overview [32]. Nearly all participant systems utilized large language models, both monolingual and multilingual. Most employed LLMs include BERT, DistilBERT, MarIA, MDEBERTa, RoBERTa, DeBERTa, Llama, and GPT-4. For processing memes, popular vision models were employed: CLIP, BEiT and ViT. Some teams employed ensembles of multiple models to enhance the overall performance. A couple of teams made use of knowledge integration to combine different language models with language features. Data augmentation techniques were used by several teams. Prompt Engineering was also used to adapt pre-trained models to the sexism detection task. Only two teams utilized deep learning architectures such as BiLSTM and CNN, while one team opted for traditional machine learning methods, including SVM, Random Forest, and XGBoost, among others. As in EXIST 2023 [31], Twitter-specific models were employed, such as Twitter-RoBERTa and Twitter-XML-RoBERTa.

While 174 systems took advantage of the multiple annotations available and provided soft outputs, 238 followed the traditional approach of providing only hard labels as outputs. Textual tasks received greater engagement, although participation is also high in the meme tasks. The binary classification tasks had more participants, followed by mono-label tasks, and finally, multi-label tasks, which is due to the increasing difficulty of these tasks.

For each of the six tasks, the organization also provided two different baseline runs: **EXIST2024 majority**, a non-informative baseline that classifies

all instances as the majority class; and **EXIST2024 MINORITY**, a non-informative baseline that classifies all instances as the minority class. The evaluation metrics for the gold standard (**EXIST2024 gold**) are also provided, in order to set the upper bound for the ICM metrics.

6 Results

In the next subsections, we report the results of the participants and the baseline systems for each task. We only show the results obtained by the best run submitted by each participant to each task. For more detailed results, please refer to the Lab Working Notes [32].

6.1 Task 1: Sexism Identification in Tweets

Soft Evaluation. Table 2 presents the results for the soft-soft evaluation for Task 1. A total of 37 runs were submitted. Out of these, 34 runs outperformed the non-informative majority class baseline (where all instances are labeled as “NO”), and all runs surpassed the non-informative minority class baseline (where all instances are labeled as “YES”). We observed a significant discrepancy in performance, with ICM-Soft Norm scores ranging from 0.6755 to 0.0374. However, if

Table 2. Results of Task 1 in the soft-soft evaluation (the best submission from each team).

Run	Rank	ICM-Soft	ICM-Soft Norm	Cross Entropy
EXIST2024 gold	0	3.1182	1.0000	0.5472
NYCU-NLP_1 [10]	1	1.0944	0.6755	0.9088
ABCD Team_3 [34]	4	0.9291	0.6490	1.2637
CIMAT-CS-NLP_3 [46]	5	0.9285	0.6489	1.2252
BAZI_1 [3]	9	0.8179	0.6311	0.9750
Awakened_2 [30]	10	0.7196	0.6154	0.8106
Victor-UNED_1 [38]	11	0.6952	0.6115	1.0691
I2C-UHU_2 [13]	13	0.6871	0.6102	0.9184
UMUTEAM_1 [28]	15	0.6679	0.6071	0.8708
MMICI_3 [15]	21	0.4589	0.5736	2.0316
clac_1	22	0.1431	0.5230	2.9543
RMIT-IR_1 [44]	23	-0.0011	0.4998	2.7892
FraunhoferSIT_1 [9]	24	-0.0658	0.4895	0.8801
CNLP-NITS-PP_1 [49]	25	-0.2086	0.4666	1.0390
Atresa-I2C-UHU_1 [6]	27	-0.3256	0.4478	3.9518
UniLeon-UniBO_1	33	-1.1882	0.3095	1.2449
EXIST2024 majority	36	-2.3585	0.1218	4.6115
NICA_3 [26]	37	-2.8848	0.0374	1.5286
EXIST2024 minority	40	-3.0717	0.0075	5.3572

we analyze the top 5 systems, we appreciate a difference of less than 5 percentual points. Notably, the best run achieved an ICM-Soft Norm score of 68% for this binary classification task, surpassing the top performance of 64% recorded by the best EXIST 2023 participant. This suggests that new models and approaches are becoming more effective at detecting sexism in social networks. However, it also indicates that there is still room for improvement.

Hard Evaluation. Table 3 presents the results for the hard-hard evaluation. In this scenario, the annotations from the six annotators are combined into a single label using the majority vote. Out of the 67 systems submitted for this task, 66 ranked above the majority class baseline (all instances labeled as “NO”). All systems surpassed the minority class baseline (all instances labeled as “YES”). Similar to the soft-soft evaluation, the results vary considerably. If we focus on the ICM-hard normalized metric, we observe that the best run gets 0.8002 while the worse one gets only 0.2665. If we focus on the top 5 systems, we observe that they achieve comparable results.

6.2 Task 2: Source Intention in Tweets

Soft Evaluation. Table 4 presents the results for the soft-soft evaluation of Task 2. The table shows that 32 runs were submitted. Among them, 25 runs achieved better results compared to the majority class baseline (where all instances are labeled as “NO”). Furthermore, all of the submitted runs outperformed or equaled the minority class baseline (where all instances are labeled as “REPORTED”). The ICM-Soft Norm scores range from the 0.4795 of the best system (“nycunlp_2”) system to 0.0000 of “fmrs_2”, indicating significant variability in the effectiveness of the submitted models. It is worth mentioning that the best system outperforms the second-best by more than 8% points. Overall, performance is considerably lower compared to Task 1. This can be attributed to the hierarchical and multiclass nature of Task 2.

Hard Evaluation. Table 5 presents the hard-hard evaluation results for Task 2, assessing 43 systems against the hard gold standard. Among these, 37 runs outperform the majority class baseline (where all instances are labeled “NO”), and all systems show equal or better performance compared to the minority class baseline (where all instances are labeled “REPORTED”). Similar to the soft-soft evaluation, discrepancies between the best and the worst-performing systems are more pronounced in Task 2 than in Task 1. The top-ranking system, “ABCD Team_1,” achieved the highest ICM-Hard normalized score (0.6320). The top 5 best systems range between 0.5937 and 0.6320. The lower end of the table includes five systems which score 0 in the ICM-Hard norm metric.

Table 3. Results of Task 1 in the hard-hard evaluation (the best submission from each team).

Run	Rank	ICM-Hard	ICM-Hard Norm	F1 _{YES}
EXIST2024 gold	0	0.9948	1.0000	1.0000
NYCU-NLP_1	1	0.5973	0.8002	0.7944
ABCD Team_1	2	0.5957	0.7994	0.7826
CIMAT-CS-NLP_2	3	0.5926	0.7978	0.7899
EquityExplorers_2 [17]	4	0.5883	0.7957	0.7775
CIMAT-GTO_3 [50]	5	0.5848	0.7939	0.7903
I2C-UHU_2	10	0.5557	0.7793	0.7733
BAZI_1	11	0.5490	0.7759	0.7755
ADITYA_3 [39]	14	0.5418	0.7723	0.7691
MMICI_3 [15]	17	0.5324	0.7676	0.7637
NICA_1	19	0.5214	0.7621	0.7642
Awakened_3	20	0.5196	0.7611	0.7652
maven_3	22	0.5015	0.7521	0.7596
Victor-UNED_3	24	0.4934	0.7480	0.7602
RMIT-IR_3	27	0.4802	0.7414	0.7548
penta-nlp_1	29	0.4779	0.7402	0.7508
fmrs_2 [48]	35	0.4398	0.7211	0.7462
clac_1	36	0.4380	0.7201	0.7376
TextMiner_2 [16]	39	0.3926	0.6973	0.7223
CAU&ITU_2 [22]	45	0.3460	0.6739	0.7024
DLRG_1	46	0.3446	0.6732	0.7085
shm2024_3 [42]	48	0.3230	0.6623	0.7044
Atresa-I2C-UHU_1	52	0.2782	0.6398	0.6899
FraunhoferSIT_1	53	0.2320	0.6166	0.6823
CNLP-NITS-PP_1	54	0.1977	0.5994	0.6762
mc-mistral_2 [43]	56	0.0614	0.5309	0.5317
UniLeon-UniBO_2	58	-0.1870	0.4060	0.4963
NIT-Patna-NLP_1	60	-0.2975	0.3505	0.5272
shm2024_2	62	-0.3410	0.3286	0.4922
DadJokers_1	63	-0.3611	0.3185	0.4365
VerbaNex_1 [23]	64	-0.4048	0.2965	0.4588
The 3 Musketeers_1 [45]	65	-0.4229	0.2875	0.3371
EXIST2024 majority	68	-0.4413	0.2782	0.0000
EXIST2024 minority	70	-0.5742	0.2114	0.5698

Table 4. Results of Task 2 in the soft-soft evaluation (the best submission from each team).

Run	Rank	ICM-Soft	ICM-Soft Norm	Cross Entropy
EXIST2024 gold	0	6.2057	1.0000	0.9128
NYCU-NLP_2	1	-0.2543	0.4795	1.8344
BAZI_1	4	-1.3468	0.3915	1.7812
Victor-UNED_2	5	-1.6440	0.3675	1.7971
ABCD Team_3	7	-1.8462	0.3513	2.4123
UMUTEAM_1	8	-1.9566	0.3424	1.4726
Awakened_2	9	-2.0091	0.3381	3.0835
fmr_1	14	-2.1737	0.3249	2.1210
CNLP-NITS-PP_1	15	-2.4732	0.3007	1.6696
Atresa-I2C-UHU_1	16	-2.6802	0.2841	2.1629
I2C-UHU_2	17	-2.6952	0.2828	2.1440
MMICI_3	20	-3.6350	0.2071	1.7285
FraunhoferSIT_1	21	-4.0856	0.1708	1.7649
RMIT-IR_1	23	-4.5481	0.1336	3.5776
CUET-SSTM_1	26	-5.1320	0.0865	4.8736
EXIST2024 majority	27	-5.4460	0.0612	4.6233
NICA_2	28	-5.7592	0.0360	2.7026
UniLeon-UniBO_3	30	-5.7633	0.0356	2.1267
EXIST2024 minority	35	-32.9552	0.0000	8.8517

6.3 Task 3: Sexism Categorization in Tweets

Soft Evaluation. Table 6 displays the results of the soft-soft evaluation for Task 3. A total of 30 runs were submitted, with 26 runs surpassing the majority class baseline (all instances labeled as “NO”), and all systems outperforming the minority class baseline (all instances labeled as “SEXUAL-VIOLENCE”). The “NYCU-NLP” team has the top three runs, with “NYCU-NLP_1” ranked first (ICM-Soft: -1.1762 , ICM-Soft Norm: 0.4379). The next two runs from the same team, “NYCU-NLP_2” and “NYCU-NLP_3,” follow closely, indicating the consistency and robustness of their approach. The fourth and fifth systems, however, show a significantly poorer performance (0.3835 and 0.3732 , respectively). The range of ICM-Soft Norm scores (from 0.4379 to 0.0000) underscores a significant variability in system performance. However, despite the complexity of the task, it seems that systems are still able to correctly capture relevant information concerning the different types of sexism.

Hard Evaluation. In the hard-hard evaluation context for the third task, 31 systems were submitted. As shown in Table 7, 28 systems outperformed

Table 5. Results of Task 2 in the hard-hard evaluation (the best submission from each team).

Run	Rank	ICM-Hard	ICM-Hard Norm	Macro F1
EXIST2024 gold	0	1.5378	1.0000	1.0000
ABCD Team_1	1	0.4059	0.6320	0.5677
NYCU-NLP_3	2	0.3522	0.6145	0.5410
CUET-SSTM_1	5	0.2883	0.5937	0.5383
CIMAT-CS-NLP_2	7	0.2643	0.5859	0.5171
penta-nlp_1	9	0.2089	0.5679	0.4856
BAZI_1	10	0.1883	0.5612	0.4843
I2C-UHU_2	11	0.1815	0.5590	0.4980
Awakened_2	12	0.1812	0.5589	0.4826
fmrs_1	14	0.1609	0.5523	0.4978
NICA_2	15	0.1506	0.5490	0.4738
RMIT-IR_1	18	0.0855	0.5278	0.4024
Victor-UNED_1	19	0.0851	0.5277	0.3257
maven_1 [40]	26	-0.0510	0.4834	0.4563
MMICI_1	27	-0.0987	0.4679	0.4548
DLRG_1	29	-0.1171	0.4619	0.3931
Atresa-I2C-UHU_1	31	-0.1524	0.4504	0.4278
CNLP-NITS-PP_1	33	-0.2694	0.4124	0.3743
FraunhoferSIT_1	34	-0.4106	0.3665	0.3823
CAU&ITU_1	35	-0.4711	0.3468	0.2998
shm2024_1	37	-0.8873	0.2115	0.3148
EXIST2024 majority	39	-0.9504	0.1910	0.1603
UniLeon-UniBO_3	41	-1.2145	0.1051	0.2605
NIT-Patna-NLP_1	42	-1.9410	0.0000	0.1207
EXIST2024 minority	46	-3.1545	0.0000	0.0280

the majority class baseline (all instances labeled as “NO”), while all systems achieved better results than the minority class baseline (all instances labeled as “SEXUAL-VIOLENCE”). The discrepancy between the best (“ABCD Team_1”, 0.5862 ICM-Hard norm score) and the worst-performing system (“CAU&ITU” 1, 0.000 score) is over 0.5 ICM-hard-norm, which is less than in Task 2. Finally, comparing the performance of the three different textual tasks in the hard-hard evaluation, the efficiency of the systems in this task, in terms of ICM-Hard Norm, is lower than in previous tasks. This further highlights the complexity of categorizing sexism.

Table 6. Results of Task 3 in the soft-soft evaluation (the best submission from each team).

Run	Rank	ICM-Soft	ICM-Soft Norm
EXIST2024 gold	0	9.4686	1.0000
NYCU-NLP_1	1	-1.1762	0.4379
Medusa_1 [2]	4	-2.2055	0.3835
ABCD Team_3	7	-3.5160	0.3143
Awakened_2	9	-4.0748	0.2848
NICA_2	12	-4.4324	0.2659
FraunhoferSIT_1	14	-5.1905	0.2259
Victor-UNED_1	15	-5.5936	0.2046
CNLP-NITS-PP_1	17	-5.7385	0.1970
RMIT-IR_1	19	-7.2098	0.1193
MMICI_3	20	-7.6413	0.0965
fmr_1	24	-8.2508	0.0643
EXIST2024 majority	28	-8.7089	0.0401
UniLeon-UniBO_1	29	-10.3622	0.0000
Atresa-I2C-UHU_1	32	-10.4052	0.0000
EXIST2024 minority	33	-46.1080	0.0000

Table 7. Results of Task 3 in the hard-hard evaluation (the best submission from each team).

Run	Rank	ICM-Hard	ICM-Hard Norm	Macro F1
EXIST2024 gold	0	2.1533	1.0000	1.0000
ABCD Team_1	1	0.3713	0.5862	0.6004
NYCU-NLP_3	3	0.3069	0.5713	0.6130
Awakened_2	6	-0.0042	0.4990	0.4833
RMIT-IR_3	8	-0.0344	0.4920	0.5049
ABCD Team_2	12	-0.1090	0.4747	0.5286
NICA_2	13	-0.2383	0.4447	0.4564
penta-nlp_1 [41]	14	-0.2597	0.4397	0.4379
maven_1	15	-0.2654	0.4384	0.4491
UniLeon-UniBO_1	16	-0.3188	0.4260	0.5032
UMUTEAM_1	20	-0.7339	0.3296	0.4942
FraunhoferSIT_1	21	-0.7437	0.3273	0.3724
MMICI_3	23	-0.8105	0.3118	0.4805
CNLP-NITS-PP_1	25	-0.9571	0.2778	0.2684
fmr_3	29	-1.5952	0.1296	0.1087
EXIST2024 majority	30	-1.5984	0.1289	0.1069
CAU&ITU_1	33	-2.3423	0.0000	0.1705
EXIST2024 minority	34	-3.1295	0.0000	0.0288

6.4 Task 4: Sexism Identification in Memes

Soft Evaluation. Table 8 presents the results for the classification of memes as sexist or not sexist. The performance results are notably low for a binary classification task: “Victor-UNED_1”, the top-ranked participant, achieved an ICM-Soft Norm score of 0.4530 and a relatively low Cross Entropy of 1.1028. However, the variability between the best and worst-performing systems is reduced compared to that of the tasks described above. When comparing these results to those of Task 1 (classifying tweets as sexist or not), we observe a significant drop in performance for image classification (0.4530 versus 0.6755 ICM-Soft Norm). It is important to highlight that most approaches relied solely on the text within the meme for classification, without incorporating image processing. This suggests that sexism in memes might often be conveyed through the imagery, even when the accompanying text seems to be neutral.

Table 8. Results of Task 4 in the soft-soft evaluation (the best submission from each team).

Run	Rank	ICM-Soft	ICM-Soft Norm	Cross Entropy
EXIST2024 gold	0	3.1107	1.0000	0.5852
Victor-UNED_1	1	-0.2925	0.4530	1.1028
Elias&Sergio_1	3	-0.3225	0.4482	0.9903
I2C-Huelva_3	4	-0.3263	0.4476	1.5189
RMIT-IR_2	8	-0.3780	0.4392	0.9852
NICA_1	9	-0.4360	0.4299	0.9278
PINK_2 [35]	10	-0.4396	0.4293	0.9375
ROCurve_3	12	-0.4646	0.4253	0.9609
the gym nerds_2	13	-0.5015	0.4194	0.9201
MMICI_2	16	-0.6183	0.4006	0.9143
OppositionalOppotision_1	21	-0.9556	0.3464	3.2025
melialo-vcassan_1	22	-1.0022	0.3389	0.9931
CNLP-NITS-PP_2	27	-1.2354	0.3014	1.0918
CHEEXIST_2	28	-1.2710	0.2957	1.1993
Penta-ML_2 [4]	30	-1.2910	0.2925	2.2277
epistemologos_1	31	-1.3486	0.2832	2.9425
EXIST2024 majority	36	-2.3568	0.1212	4.4015
EXIST2024 minority	38	-3.5089	0.0000	5.5672

Hard Evaluation. Table 9 presents the results for the hard-hard evaluation of Task 4. Out of the 50 systems submitted for this task, only 37 ranked above

Table 9. Results of Task 4 in the hard-hard evaluation (the best submission from each team).

Run	Rank	ICM-Hard	ICM-Hard Norm	F1 _{YES}
EXIST2024 gold	0	0.9832	1.0000	1.0000
RoJiNG-CL_3 [20]	1	0.3182	0.6618	0.7642
I2C-Huelva_2	4	0.1313	0.5668	0.7241
DiTana-PV_2 [25]	6	0.1150	0.5585	0.7122
Victor-UNED_2	7	0.1028	0.5523	0.7154
MMICI_2	8	0.1014	0.5515	0.7261
I2C-Huelva_3	9	0.0987	0.5502	0.6933
NICA_1	11	0.0767	0.5390	0.7248
OppositionalOppotision_1	14	0.0494	0.5251	0.7168
Elias&Sergio_1	15	0.0433	0.5220	0.6979
ROCurve_3	19	0.0088	0.5045	0.6834
PINK_1	20	0.0076	0.5039	0.7044
RMIT-IR_2	22	-0.0123	0.4938	0.6726
Miqarn_1	26	-0.1159	0.4411	0.6632
CNLP-NITS-PP_1	27	-0.1234	0.4372	0.6699
Penta-ML_2	28	-0.1308	0.4335	0.6742
epistemologos_1	30	-0.1823	0.4073	0.5503
TokoAI_1	31	-0.1872	0.4048	0.5639
UMUTEAM_1	33	-0.2422	0.3768	0.6963
Umera Wajeed Pasha_1 [29]	36	-0.3083	0.3432	0.5956
TargaMarhuenda_1	37	-0.3535	0.3202	0.6487
EXIST2024 majority	39	-0.4038	0.2947	0.6821
DLRG_1	41	-0.4206	0.2861	0.6469
MIND_1 [21]	42	-0.4986	0.2465	0.5674
ALC-UPV-JD-2_1	43	-0.5446	0.2231	0.4878
dap-upv_1 [27]	44	-0.5737	0.2082	0.4188
AI Fusion_1	45	-0.6416	0.1737	0.4651
EXIST2024 minority	46	-0.6468	0.1711	0.0000
TheATeam_1	50	-0.6644	0.1621	0.4821
melialo-vcassan_2	51	-0.6644	0.1621	0.0281

the majority class baseline (all instances labeled as “NO”), while 47 systems surpassed the minority class baseline (all instances labeled as “YES”). Similar to the soft-soft evaluation, the results vary considerably, from 0.6618 ICM-Hard Norm for the best performing system (RoJiNG-CL_3) to 0.0876 (melialo-vcassan_1).

6.5 Task 5: Source Intention in Memes

Soft Evaluation. Table 10 presents the results for the classification of memes according to the intention of the author, with the outputs provided as the probabilities of the different classes. Only 15 runs were submitted for this task. While all the runs ranked above the minority class baseline (all instances labeled as “JUDGEMENTAL”), only 15 runs surpassed the majority class baseline (all instances labeled as “NO”). The results for this task are notably low, with the best team (“Victor-UNED_2”) achieving only 0.3676 ICM-Soft Norm. This suggests that identifying whether a meme contains direct sexism or is judgmental is more difficult than identifying the intention behind a sexist tweet.

Table 10. Results of Task 5 in the soft-soft evaluation (the best submission from each team).

Run	Rank	ICM-Soft	ICM-Soft Norm	Cross Entropy
EXIST2024 gold	0	4.7018	1.0000	0.9325
Victor-UNED_2	1	-1.2453	0.3676	1.6235
MMICI_1	2	-1.2660	0.3654	1.4645
NICA_1	4	-1.5329	0.3370	1.4664
CNLP-NITS-PP_1	5	-1.5907	0.3308	1.5273
melialo-vcassan_2	6	-1.9847	0.2889	1.5211
I2C-Huelva_3	10	-2.7996	0.2023	3.9604
EXIST2024 majority	14	-5.0745	0.0000	5.5565
Penta-ML_3	15	-5.2668	0.0000	5.1547
EXIST2024 minority	18	-18.9382	0.0000	8.0245

Hard Evaluation. Table 11 presents the results for the hard-hard evaluation of Task 5. Out of the 19 systems submitted for this task, only 15 ranked above the majority class baseline (all instances labeled as “NO”), while 18 systems surpassed the minority class baseline (all instances labeled as “JUDGEMENTAL”). The results range from 0.4167 ICM-Hard Norm for the best performing system (“Victor-UNED_1”) to 0.0000 for the worst performing systems, but are quite homogeneous among the top 5 systems.

Table 11. Results of Task 5 in the hard-hard evaluation (the best submission from each team).

Run	Rank	ICM-Hard	ICM-Hard Norm	Macro F1
EXIST2024 gold	0	1.4383	1.0000	1.0000
Victor-UNED_1	1	-0.2397	0.4167	0.3873
I2C-Huelva_2	2	-0.2535	0.4119	0.4761
NICA_1	6	-0.2881	0.3999	0.3837
MMICI_1	7	-0.3066	0.3934	0.4179
CNLP-NITS-PP_1	9	-0.3370	0.3829	0.4101
Penta-ML_3	11	-0.6123	0.2872	0.3841
TokoAI_1	14	-0.7263	0.2475	0.3716
melialo-vcassan_3	15	-0.7758	0.2303	0.3709
EXIST2024 majority	17	-1.0445	0.1369	0.1839
UMUTEAM_1	18	-1.1486	0.1007	0.2098
DLRG_1	20	-1.4891	0.0000	0.2530
EXIST2024 minority	21	-2.0637	0.0000	0.0697
epistemologos_1	22	-8.7012	0.0000	0.0557

6.6 Task 6: Sexism Categorization in Memes

Soft Evaluation. Table 12 presents the results for classifying memes based on the aspects of women being attacked, with outputs provided as class probabilities. Only 19 runs were submitted for this task. While all runs performed better than the minority class baseline (labeling all instances as “MISOGYNY-NON-SEXUAL-VIOLENCE”), only 11 runs exceeded the majority class baseline (labeling all instances as “NO”). The performance for this task was generally low, with the top team (“ROCurve_1”) achieving an ICM-Soft Norm score of only 0.2462, which is significantly lower compared to the results for the same task when applied to tweets (Task 3).

Hard Evaluation. Finally, Table 13 presents the results for classifying memes based on the aspects of women being attacked, with outputs provided as a single class prediction. 22 runs were submitted for this task. Only 17 runs exceeded the majority class baseline (labeling all instances as “NO”), while 21 runs ranked above the minority class (all instances labeled as “MISOGYNY-NON-SEXUAL-VIOLENCE”) The performance for this task was low, with the top team (“DiTana-PV_1”) achieving an ICM-Soft Norm score of 0.3549.

Table 12. Results of Task 6 in the soft-soft evaluation (the best submission from each team).

Run	Rank	ICM-Soft	ICM-Soft Norm
EXIST2024 gold	0	9.4343	1.0000
ROCurve_1	1	-4.7893	0.2462
the gym nerds_2	2	-4.7942	0.2459
Elias&Sergio_1	5	-5.9160	0.1865
Victor-UNED_1	6	-6.4124	0.1602
CNLP-NITS-PP_1	8	-6.6782	0.1461
AI Fusion_1	10	-7.6282	0.0957
EXIST2024 majority	13	-9.8173	0.0000
dap-upv_1	14	-10.4213	0.0000
Penta-ML_2	15	-11.2593	0.0000
MMICI_1	19	-16.1248	0.0000
EXIST2024 minority	22	-50.0353	0.0000

Table 13. Results of Task 6 in the hard-hard evaluation (the best submission from each team).

Run	Rank	ICM-Hard	ICM-Hard Norm	Macro F1
EXIST2024 gold	0	2.4100	1.0000	1.0000
DiTana-PV_1	1	-0.6996	0.3549	0.4319
MMICI_1	3	-0.9863	0.2954	0.4342
ROCurve_1	4	-1.0089	0.2907	0.3639
Penta-ML_3	8	-1.3631	0.2172	0.3356
Elias&Sergio_1	11	-1.5276	0.1831	0.4321
Miqarn_1	13	-1.6216	0.1636	0.3211
CNLP-NITS-PP_1	14	-1.7920	0.1282	0.1587
ALC-UPV-JD-2_1	15	-1.8573	0.1147	0.2103
dap-upv_1	17	-1.9497	0.0955	0.2227
UMUTEAM_1	18	-1.9511	0.0952	0.3786
EXIST2024 majority	19	-2.0711	0.0703	0.0919
TargaMarhuenda_1	20	-2.0725	0.0700	0.1440
TheATeam_1	22	-2.3159	0.0195	0.1490
EXIST2024 minority	23	-3.3135	0.0000	0.0318
One-by-zero_1	25	-4.5910	0.0000	0.2304

7 Conclusions

The objective of the EXIST challenge is to encourage research on the automated detection and modeling of sexism in online environments, with a specific focus on social networks. The EXIST 2024 Lab held as part of CLEF attracted nearly 60 participant teams, and received a total of 412 runs. Participants adopted a wide range of approaches, including vision transformer models, data augmentation through automatic translation, data duplication, utilization of data from past EXIST editions, multilingual language models, Twitter-specific language models, and transfer learning techniques from domains like hate speech, toxicity, and sentiment analysis. While many systems opted for the traditional approach of providing only hard labels as outputs, a significant number of systems leveraged the multiple annotations available in the dataset, and provided soft outputs, proving that there is an increasing interest by the research community in developing systems able to deal with disagreements and with different perspectives.

For future editions of EXIST, we plan to expand our study in order to include additional communication channels and media formats, such as TikTok videos. By doing so, we aim to address the nuances and unique challenges presented by different formats, enhancing the robustness and applicability of research on automated sexism detection. Additionally, this expansion will allow us to capture a broader spectrum of online interactions and cultural contexts.

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